

Aerospace Engineering Portfolio

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The University of Arizona: Department of Engineering

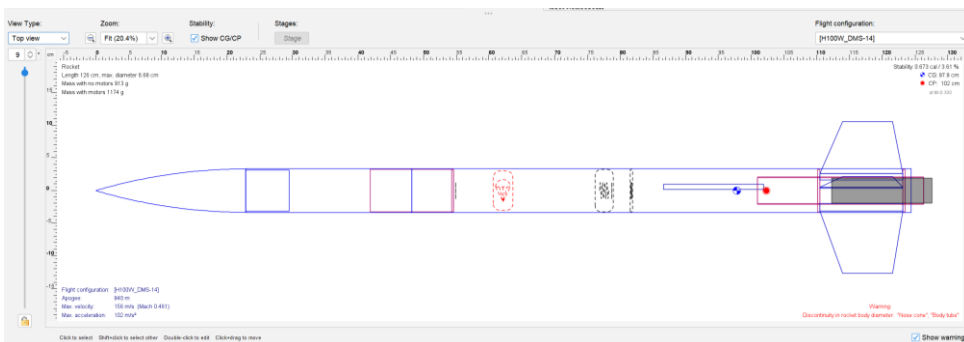
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Level 1 Rocket; American Institute of Aeronautics and Astronautics



What?

The purpose of this rocket was to demonstrate knowledge of composite material applications, rocket design, and safety operations. The successful flight and recovery of this rocket earned me my Level 1 Certification through the National Association of Rocketry (NAR), allowing me to purchase “H” through “I” class rocket motors (up to 640 N·s total impulse).

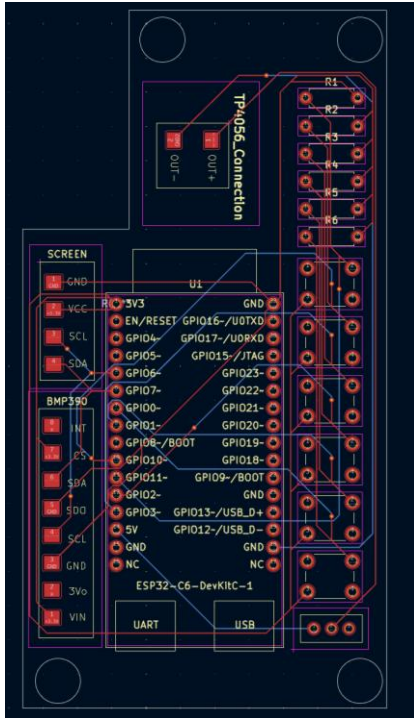
How?

- **Modeling and Simulation**
Using the modeling software OpenRocket, I was able to model my rocket to determine the center of gravity, center of pressure, and the expected altitude the rocket would achieve (See figure above).
- **Hardware**
The rocket, the HI-TECH PK-56, came with most of the parts required to build the rocket. AIAA provided a custom fiberglass and epoxy mix to secure parts together inside the rocket to prevent damage and breakage during flight.

Results

The rocket launched, flew, deployed its parachute, and returned safely, earning me Level 1 Certification. The flight carried no onboard altimeter, so the actual apogee could not be compared against the OpenRocket prediction; this is the gap that motivated my next project.

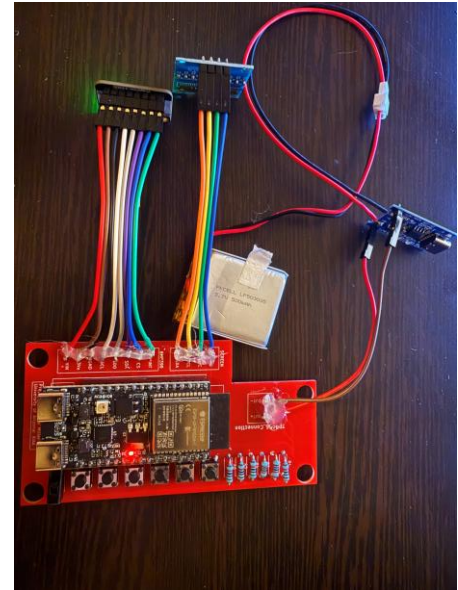
Altimeter and Level 2 Rocket; American Institute of Aeronautics and Astronautics



How?

Altimeter

▪ **Modeling and Hardware**
Controlled by an ESP32-C6-DevKitC-1 microcontroller running MicroPython, this circuit uses a BMP390 barometric pressure sensor to determine and save the system's height over multiple launches. The maximum altitude and a graph of altitude vs. time are then displayed for easy retrieval of results. After breadboard prototyping, I designed a two-layer PCB in KiCad and had it fabricated by JLCPCB.



Results and Expectations

Altimeter

The altimeter is capable of recording and displaying the height achieved. The BMP390 is able to record height changes in 0.1 meters and can be tested by hand for accurate recordings. Both display settings work as intended.

Level 2 Rocket

OpenRocket will be used to determine the projected altitude. The altimeter will be placed inside and must be able to withstand flight and landing so the achieved altitude can be properly acquired.

What?

Altimeter

The goal of the altimeter is to record the altitude the Level 2 rocket reaches. I can then compare this to the expected altitude calculated using OpenRocket.

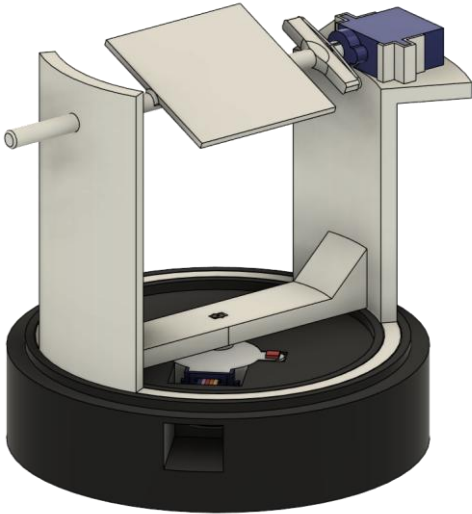
Level 2 Rocket

Being built from scratch, this rocket will use a “J”, “K”, or “L” class motor to reach an expected apogee (which will be determined after it is built) to earn my Level 2 Certification.

▪ **Coding**

I implemented the MicroPython firmware on the ESP32 across four modules: a custom BMP390 I²C driver, OLED rendering, LittleFS flash storage, and the main control loop. I designed the module structure and used AI assistance to speed up implementation.

Dual Axis Solar Tracker



How?

▪ Modeling and Simulation

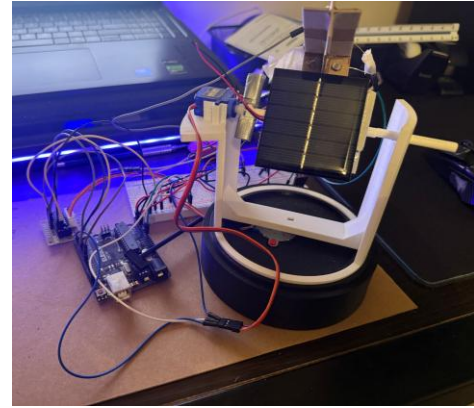
Using Autodesk Fusion 360, I created a 3D assembly of our project. Our model included the placement of the motors along each axis of rotation to accurately visualize our expected final product.

▪ Hardware

We used an Arduino Uno microcontroller as our main control. We used four light-dependent resistors (LDRs), placed hidden from each other, to guide the motors to turn the solar panel toward the sun.

▪ Coding

We defined the control logic and LDR comparison thresholds, used AI assistance to generate the initial C++ firmware, then debugged until tracking was stable without frequent oscillation.



Results

▪ Observations

Our solar tracker was capable of rotating the solar panel directly toward the sun. Future improvements would be made to deliver more power to the stepper motor to increase the speed of rotation.

▪ Data

Voltage, current, and power were measured against light intensity using a multimeter and lux meter. All three rose with irradiance, confirming the tracker's advantage over a fixed panel. Measured efficiency was 0.77%, far below typical silicon, because voltage and current were recorded separately rather than at a single loaded operating point.

What?

The goal of this project was to create a dual-axis solar tracker to maximize the amount of solar energy a small solar panel could absorb.

As a four person team, I was responsible for 3D modeling the structure that the solar tracker would be built around. I also helped wire the microcontroller to the sensors and motors and assisted with the required programming.

Undergraduate Courses

Unweighted GPA, Honor Awards, and Graduation Details

GPA: 3.903

Dean's List with Distinction: Full Time: Issued 12/18/2025

Dean's List: Full Time: Issued 05/14/2026

Expected Graduation: Spring 2029

Completed Courses: Spring 2026

AME 209: Computer Programming in Aerospace and Mechanical Engineering

- **MATLAB** simulation experience.
- **Python** simulation experience.

Currently Enrolled: Fall 2026

AME 211: Computer-Aided Drafting and Manufacturing

- Experience in hand-drawing blueprints.
- Experience with **SolidWorks** and **AutoCAD**.

AME 220: Introduction to Aerospace Engineering

- Aerodynamics fundamentals: continuity, momentum, and energy equations; compressible flow and shock waves.
- Standard atmosphere modeling: pressure, density, and geopotential altitude relations.
- Airfoil and finite-wing analysis; lift, drag, and aircraft performance.
- Orbital mechanics: Kepler's laws, the vis-viva equation, and Hohmann transfer orbits.

Experience

Teaching

MATH 391: Vector Calculus Undergraduate Learning Assistant: *Began Fall 2026 – Current*

- Lead drop-in tutoring sessions for enrolled students.
- Assist students with calculus material during class.
- Grade students' completed work.
- Fall 2026 Class Size: 89

Engineering and Leadership

American Institute of Aeronautics and Astronautics: University of Arizona: *Fall 2025 – Current*

- Computer-Aided Design (CAD): 3D modeling and PCB design in **KiCad**.
- Experience with rocket simulation and design with **OpenRocket**.
- Construction experience with composite materials.
- Participate in safety and launch operations.
- Earned HPR Level 1 Certification through the National Association of Rocketry:
Issued 07/09/26

Eagle Scout: Troop 17, Jacksonville, OR: Bronze, Silver, and Gold Eagle Palms.

- Project: Planned and led a volunteer team to design and install trail signage on a local hiking trail while coordinating materials and landowner approval

Robotics Team: St. Mary's School, Medford, OR: 09/2022 – 06/2025

- Designed and prototyped competition assemblies in **Fusion 360** and **SolidWorks**.
- Operated shop machinery and 3D printers to manufacture parts from CAD models.

Work

Maintenance Painter: Medford School District, Medford, OR: 06/2026 – 08/2026

- Completed surface preparation and repainting across elementary school sites on a fixed summer schedule.
- Applied equipment, ladder, and occupied-facility safety procedures daily.

Server and Shift Support: Platon's Kitchen, Medford, OR: 06/2022 – 07/2025

- Three years of continuous employment through high school while maintaining a 3.97 GPA.
- Coordinated front-of-house staffing and service flow during peak hours.